Carla - Maria Bărăștean

Automation and Computer Science Graduate

Contact

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Location: Cluj-Napoca, Romania
Portfolio: website
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Profile

Automation and Computer Science graduate focused on nonlinear control, model-based development and applied AI.

I build reproducible engineering software, from MATLAB toolboxes and embedded-control models to full-stack applications.

Focus

- Control Systems
- Software Engineering
- Applied AI

Hard skills

Control
MATLAB, Simulink, ASCET, TPT

Programming
Python, C/C++, JavaScript, TypeScript

AI / Vision
TensorFlow, PyTorch, OpenCV

Web / Data
React, Node.js, MongoDB

Tools
Git, Docker, STM32

Soft skills

- Leadership
- Analytical thinking
- Adaptability
- Communication
- Attention to detail

Languages

Romanian	(Native)
English	(C1)
German	(A2)
Italian	(A2)

Experience

Model-Based Software Developer
Jul 2024 - Jan 2025
Bosch, Cluj-Napoca

Built and validated a six-component electric-vehicle drivetrain model in Simulink, ASCET and TPT. Generated production C code for STM32 controllers and developed a monitoring interface.

Computer and Network Operator
2019 - Present
ALBALUX COM

Install and configure hardware and networks; diagnose workstation and connectivity issues; provide on-site and remote technical support.

Education

B.Sc. Automation and Computer Science
2022 - 2026
Technical University of Cluj-Napoca

Graduated with a final average of **9.25**/10.
Thesis: *Automation of Nonlinear Control Analysis and Design: A MATLAB Implementation Based on Feedback Linearization.*

Decebal National College
2018 - 2022
Deva, Romania

Mathematics and Computer Science in English

Research

As the Crow Flies, or as the River Flows?
2026
Co-author - manuscript submitted to ICCP 2026

Graph Topology vs. Proximity in Satellite Water-Quality Monitoring.
Controlled comparison of physical river-flow topology and geographic proximity for satellite water-quality monitoring across Germany and Romania.

Selected Projects

EcoSort - Intelligent Waste Recognition
2026
Python, TensorFlow, OpenCV, Arduino

Real-time material classification and optional Arduino-controlled physical sorting.

BudgetLy - Collaborative Finance Platform
2025
React, Node.js, MongoDB, Express

Full-stack platform for financial analytics, recurring transactions and shared-payment tracking.

Selected Activities

CASSINI Hackathon - AquaGraph
2026
3rd in Romania, 1st at the Cluj-Napoca hub

Developed a space-data solution for monitoring and preventing river pollution.

Panel Speaker - IEEE SACI 2025
2025
ADAPTED Research Group

Discussed innovation in STEM and academic entrepreneurship at an international conference workshop.